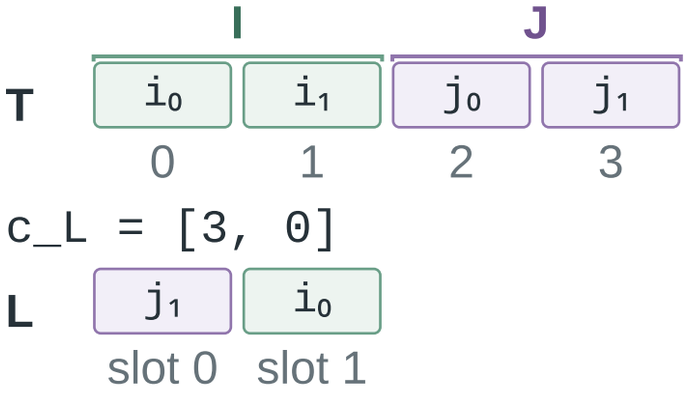
`dot(dot(X, Y), Z) → dot(X, dot(Y, Z))`

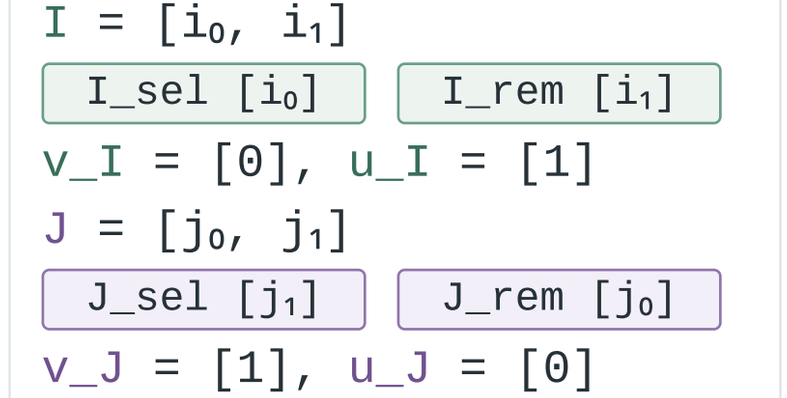
T = dot(X, Y), O = dot(T, Z). Inner dot contracts K; outer dot contracts L.

1. Select axes



Tensor order is unchanged

2. Partition by origin



Views retain axis positions

3. Preserve pairings

Physical Z axes



c_R = [2, 0]

Slot	From T		From Z	Z pos
0	j_1	↔	z_1	2
1	i_0	↔	z_0	0

Selected subgroup mappings

Positions are zero-based

Selected group	Producer positions	Slots in L	Partner positions in Z
I_sel	v_I	u_I	gather(c_R, u_I)
J_sel	v_J	u_J	gather(c_R, u_J)

U = dot(Y, Z) contracts J_sel; final dot contracts I_sel + K.

U retains K and I_sel's partners. Output: I_rem ++ J_rem ++ F (F = [f]).